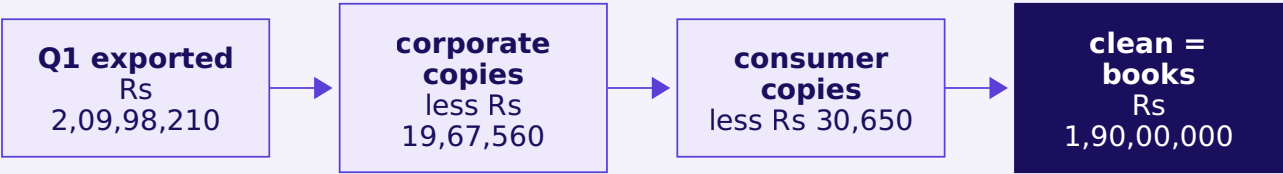


Which Q1 figure is right, and how do we know?

Kalpa Retail's dashboard reads Q1 at Rs 2.1 crore from an orders CSV out of the ERP, the enterprise resource planning system Finance books orders in, stitched from two extracts, two pulls of rows. The books, Finance's own record of Q1, say Rs 1.9 crore. Two honest totals disagree until a bridge walks one to the other, one move per cause, and every cleaning act on the way is a decision with a reason.

1 Does the bridge from Rs 2.1 crore close to the books?



Yes, in rows and in rupees. In rows, 201 in equals 186 kept plus 15 set aside, each removed with a logged reason. In rupees, Q1 as exported less the rupees set aside equals the books to the rupee.

Crux. Reconcile twice, in rows and in rupees, to the books.

2 In what order does the cleaning pass run?

Step	What it does	The failure it prevents
Profile	Present, convertible, distinct per field	A total from an unread file
Identity rule	One row per order_id, keeping the copy whose amount converts	Copies in revenue
Convert	Value or reason, each failure logged	An order worth Rs 0
Keep, drop or flag	One decision per defect, with a reason	A guess that becomes a fact
Reconcile	Rows and rupees, to the books	A pass that rounds to the books
Recompute	Every number already reported	A finding nobody rechecked

Crux. Profile before you total: present, convertible, distinct, for every field.

3 How do you convert text without inventing a number?

```
def convert(value):
    try:
        return int(value), ""
    except (TypeError, ValueError):
        return None, "amount does not convert"
```

Everything read from a CSV is text, so "900" < "1200" is False. A missing value is "" in a CSV and an absent key in JSON.

Crux. A failure is logged, never turned into a number; large is not wrong.

4 When is a row a repeat, and which copy stays?

The pair	Keep	Log
Identical	The first	Second copy of the order
One amount unreadable	The copy that validates	Its twin carries the value
Valid, a field disagrees	The first extract	The field, and a question for the source

Choose the key before counting: the whole record, which carries each row's line in the file, flags 0 rows, the record less that line 13, order_id 15, and a fuzzy match on customer and amount within 60 days 15, with one real Rs 17,71,000 order among them.

Crux. Say what makes two rows one order before you count duplicates.

5 What happens to a missing value or a huge order?

Decision	Revenue	A status count	Use it when
Drop	Moves	Changes its denominator	The record is not an order
Default	Unchanged	Invents a value	A stated rule covers it
Keep and flag	Unchanged	Leaves it out	The value is unknown

A large order is a question about its record: a real order at Rs 29,45,460 in the Business segment, Kalpa's sales to companies, stays, flagged, and Q2 is shown both ways. Repair a value only from a copy that could not share the error.

Crux. Keep the copy that validates, and log every row you set aside.

6 What catches each of the day's six wrong numbers?

The number	The step	The check
Largest Q2 order Rs 970	Sorting amounts as text	Below every Business order
0 duplicates	The file line in the key	201 rows, 186 order ids
188 orders, Q2 Rs 1,87,03,710	The record less its line ties Q1, and the pass stops	Rows against ids
201 of 201 convert, an order at Rs 0	Failures turned into 0	The smallest real order is Rs 680
Q2 Rs 1,57,54,540, a 17.1% fall	The largest Q2 order removed as an outlier	A known account, valid fields
201 = 185 + 16, Rs 20,00,000 set aside	Keep the first copy, then convert	Rs 1,790 short of the books

7 What did cleaning change in Tuesday's finding?

Number	As Tuesday reported	On clean data
Revenue, Q1 to Q2	-11.0%	-1.6%
Orders per customer	x0.754	x0.860
Orders per customer in Retail-Plus, Kalpa's paid membership tier	-49.0%	-35.0%, 1.82 to 1.18

Monday's revenue tree multiplies customers, orders per customer and revenue per order. Customers stay at 69, so the fall sits in orders per customer, and it is smaller.

Crux. Recompute what you reported, and say what changed, the smaller number first.

THE WORDS ON THIS SHEET

Profile Three counts per field before any total: present, convertible, distinct

Identity rule What makes two rows the same thing

Set aside Removed from the clean file with a logged reason and the row that stayed

Keep and flag Keep a record whose value is unknown, marked, out of counts that need it

Outlier A value far from the rest; a question about its record

Reconciliation Proof the clean data is the same data, in rows and in rupees